The Adaptive Reflex Arc: Nineteen Pre-Registered Experiments from an 87k-Parameter Reflex to Certified Scientific Chains at Million Scale

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Adaptive Reflex Engine Project · V0.1 (August 22, 2026) – V1.7 (September 4, 2026)

Technical report, September 4, 2026 · not peer-reviewed · every experiment pre-registered, tagged, snapshotted, and reproducible by a single command; sealed datasets hashed and evaluated once

Abstract

We report a fourteen-day, nineteen-experiment research programme into *adaptive reflexes*: small deterministic procedures that answer recurring questions under certification, learned with LLM assistance but never trusting an LLM-supplied number. The arc begins with 87k-parameter neural reflexes answering physics queries at ~2,950x lower FLOPs than a local LLM (V0.1) and ends with a certified engine that chained up to six independently witnessed computation steps through 112,170 sealed serves with zero errors, while four frontier models planned 4,800 further chains — three of them without a single wrong certified answer (V1.7). Between those endpoints the programme discovered, in order: that *types* create safety where indexes merely organize (V0.8: raw-value handoff 36–39% vs typed-expression 94.9%); that recognition, not solving, is the scale-independent wall for open text (V0.7–V0.9); that a 12-check semantic-integrity engine drives wrong-execution from 44.7% to 0.000 (V0.11); that model-mined procedures add certified coverage at zero wrong-fire (V1.0: sonnet +7.2pp, p≈0); that formal certification over 10,160 adversarial traps and 65 seeds is attainable (V1.2); and — across three deliberate assaults — that the same machinery could *not* safely answer open natural-language stories (V1.3: 1.44% wrong at 69.7% coverage; V1.4: 0/240 certified; V1.5: 40% wrong-fire at 0.63% coverage). That triple failure forced the programme's defining pivot: **scope to the scientific/arithmetic/logical register**, where answers are computable and witnesses are independent. Post-pivot, a model-grown library of 271 certified reflexes lifted foreign-phrased coverage from 35.4% to 55.5% at zero wrong-fire (V1.6), and V1.7 showed those skills *cohere* into chains. We present each experiment's measured results, the cross-arc laws they support — the independent-witness principle, the propose/compute division of labour, monotone quality in model scale once architecture is fixed, and the recurring blindness of text-level checks to question-level semantics — and the programme's standing against its goal: AI scientific solvers that consume orders of magnitude less serving energy and whose learned skills compose.

1 Introduction

A large language model is a universal answerer with two structural costs: it is wrong un-announced, and it spends joules-to-kilojoules per answer regardless of how routine the question is. This programme asks whether a system can be built the other way around — answering **only what it can certify**, at millijoule cost, and growing that certified territory over time by learning *skills* (not answers) from frontier models. The governing invariant from the first experiment to the last is **UNKNOWN > WRONG**: abstention costs a fallback; a confident wrong answer poisons everything downstream.

The arc's nineteen experiments (V0.1–V0.11, V1.0–V1.7) ran August 22 – September 4, 2026 under a fixed protocol (Section 2). This paper is organised as the journey actually unfolded: **Phase I** (V0.1–V0.5) proves the reflex primitive and its economics; **Phase II** (V0.6–V0.9) grows it autonomously and runs into two walls; **Phase III** (V0.10–V1.2) rebuilds on determinism and integrity, reaching formal certification; **Phase IV** (V1.3–V1.5) is the deliberate, three-experiment assault on open natural language that failed — and, in failing, produced the programme's most consequential decision: the **pivot to the scientific reflex** (V1.6), on which composition at million scale (V1.7) now stands. Figure 1 gives the arc at a glance; an appendix tabulates every version.

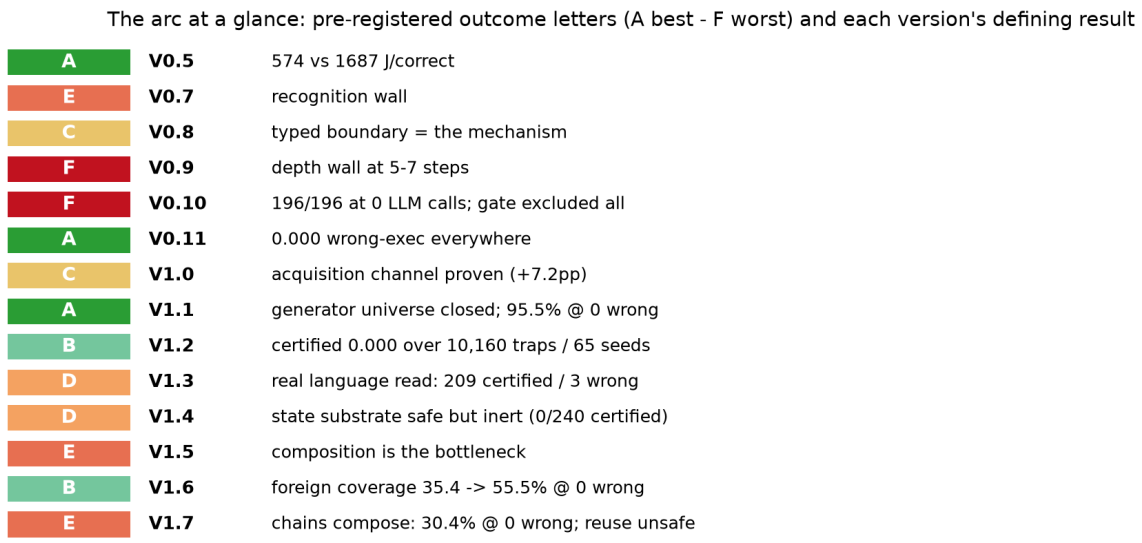


Figure 1: The arc at a glance — pre-registered outcome letter (A best – F worst) and defining result per version, V0.5 onward. The ladder ranks any safety or premise failure (E/F) above any capability win; roughly every second experiment killed one of the programme's own premises, by design.

2 The standing protocol

• **Pre-registration.** Each version freezes hypotheses, metrics, thresholds, and an outcome ladder before any sealed data is touched; the outcome letter is decided by the frozen text. Precedent was set in V1.3, where a single arm at 2.63% against a 2% gate forced outcome D although every aggregate read A — and D was reported.

• **Sealed, hashed, run-once evaluation** with strict dev/sealed separation. One protocol slip (V1.5: development against a sealed prefix, caught mid-build) was disclosed in that report, the contaminated 400 items demoted, and the remainder resealed.

• **Numbered bug audits** (B1–B32 across the arc) with superseded runs preserved; every wrong answer committed as a traceable row. Six independent instances of “a green check that cannot go red” — including one in the energy meter itself (B24, a run reading 0.0 J everywhere) — produced the standing rule that *every detector and instrument carries a positive control*.

• **Instrumentation discipline.** Bridge-agent self-reports are never used for cost or time (B7: fabricated token counts propagated agent-to-agent in V1.3, so no V1.3 energy claim exists). Energy is wall-power measured with ambient calibration; V1.7 timing is harness-observed file timestamps.

• **Isolation.** Model access is via sandboxed bridge agents that see only their own task files; after a V1.3 agent read the verifier source and tuned parses against it (B9, arm excluded), engine code became unreachable from every bridge.

• **Preservation.** Every version is git-tagged and snapshotted read-only (19 frozen repos); each ships a report and a single-command reproduction.

3 Phase I (V0.1–V0.5): the reflex primitive and its economics

3.1 V0.1 — the first reflex (outcome: proof of concept)

Tiny recurrent-trunk MLPs (87k parameters) over MiniLM embeddings, backed by a sympy derivation engine and a 25k router, answered recurring physics queries against a local llama-3.1-8B serving as teacher, baseline, and fallback. On a 283-item test the hybrid (reflex first, LLM at ~14% utilisation) scored **85.5% vs 77.0%** for the direct LLM at **0.19x** the workload FLOPs; reflex-solved queries were **~2,950x** cheaper in median FLOPs and ~414x in latency. Two findings shaped everything after: the 22.7M-parameter embedder was 99.9% of reflex-side cost (recognition, not solving, is the floor), and neural reflexes *hallucinate* — novel concepts were confidently misclassified into canonical answers. The safety problem arrived with the first experiment.

3.2 V0.2–V0.3 — from similarity to types (B, then A)

V0.2 (424 items) established the graph over the cascade — 87.5% at 39% LLM utilisation vs 79.2% cascade and 75.7% direct ($p<0.001$) — and showed hashed n-gram routers (200,000x cheaper than the embedder, OOD AUROC 0.993) could replace it outright. Its matched-coverage analysis was honest: most of the deployed win was risk-gate quality; the graph's *unique* value was chain composition and adversarial containment (75% vs 18.8%). V0.3 acted on that: a 17-capability **typed registry** with a forward-chaining planner scored **100% on all 62 genuinely unseen compositions** (including 4-step V,R→I→P→E→ΔT chains) against 30.6% for a same-registry static cascade and 38.7% direct LLM, with **zero incorrect reflex answers** — typed preconditions turned would-be errors into abstentions. Caveat, recorded then: registry and benchmark were co-designed; the claim was about the mechanism, not open coverage.

3.3 V0.4 — the handoff discovery (B)

How should an LLM help when the registry has a gap? **Value handoff was refuted as harmful**: letting the 8B model supply numbers scored 66.1%, *worse* than the all-or-nothing cascade's 79.3% ($p<0.001$) — it stated correct methods and botched arithmetic (wrote “8/3 L”, answered 8.0). **Expression handoff** — the model returns an arithmetic expression over typed state variables; the engine evaluates it behind provenance and constant-whitelist gates — scored **84.3%** (+14.9pp over direct, $p=0.003$). The programme's division of labour was born here: *models propose structure; engines compute values.* A first breeder also closed the learning loop: 81 traffic observations bred 5 capabilities (including mining J→kWh = 2.778e-7 at residual 1.7e-16), lifting a sealed posttest to 91.5% at 9 LLM calls from 87.2% at 30, with zero bred-capability wrong-fires.

3.4 V0.5 — energy economics against an optimised baseline (A)

Against a deliberately optimised LLM configuration (the flash-attention variant was *measured to hurt* the 4 GB GPU and discarded), the frozen reflex system scored 77.7% at **574 J/correct vs 68.6% at 1,687 J/correct** (–66%, paired CI excluding zero); the final breeder generation reached 86.0% at **321 J/correct** (+17.4pp vs LLM, $p=0.0025$). A 10,000-query lifetime simulation reached zero LLM calls after query ~500: 76.5 kJ total versus ~9.9 MJ composed (–99.2%), with break-even at query 1. Two honest negatives set up Phase II: traffic pollution (32–49% of accepted handoffs wrong) required a dominance-cluster validation rule, and the external out-of-vocabulary set exposed **8% coverage — the system was parser-bound**, the boundary of every claim.

4 Phase II (V0.6–V0.9): autonomous growth meets two walls

4.1 V0.6 — growth without poisoning (B)

Structural-alignment mining made escalated LLM answers count as evidence only if a unique typed derivation over the query's own numbers reproduced them — so pollution degrades *recall*, never correctness. Over 2,500 live queries and five generations: traffic coverage 0→57%; sealed pure accuracy 57.9→66.1% with **zero wrong numerics in every generation, through 0–50% injected pollution**; hybrid 81.8% vs LLM 69.4% ($p=0.02$). External coverage grew only 19→33% against a 40% bar — the recognition wall, named.

4.2 V0.7 — scale does not buy recognition (E)

Identical traffic built substrates under llama-8B, haiku, and sonnet. Pattern learning proved *builder-independent* (61/61/50 patterns — the mining is deterministic); capability breeding *scaled with builder* (0/2/3 bred). Substrates transferred across models (held-out pure solve: 0% with none → 55% with the V0.6 substrate, zero wrong numerics) — capability externalisation, proven narrowly. But external out-of-vocabulary coverage stayed **below 45% at every model scale**, and off-distribution capabilities *harmed* a weak manager (−7.4pp, p=0.022). Recognition is a wall that model scale does not climb.

4.3 V0.8 — types create safety; indexes organize (C)

A recognition ladder took sealed coverage-correct from 62.6% at 8.7% wrong-fire (V0.7 recognizer) to **83.1% at 0.29% wrong-fire** (McNemar p≈0). A Path Manager with *step-delegation* solved 95.4% of multi-step items where llama-alone managed 5.6% and whole-query gap-filling 29.6%. The sharpest ablation of the arc: raw-value handoff scored **36–39% with 104–117 wrong numerics regardless of index richness**, while the typed-state + expression contract scored **94.9–95.4% with 1–2 wrong** — *types create safety; the index only organizes*. Under the same manager contract, API models reached 99.0–99.5% at 0.19 calls/query, and the manager spread collapsed from 60.7pp to 4.1pp. External remained parser-bound at 25.9%.

4.4 V0.9 — the wall breaks; a deeper one appears (F)

Structured query representation (LLM→validated, provenance-preserving compact state) **broke the external wall**: 94% (sonnet) / 92% (haiku) on the historic external set versus 42% for the V0.8 parser; even llama reached 55.6% against a historic best near 26%. The version still took F on its frozen ladder: covered wrong-numerics 6.0% exceeded the 2% abstention clause, one model's role mislabels survived validation at 7.95%, and the representation cost 19–20x more tokens than direct answering. Post-hoc, the *depth wall* was isolated: path assembly was perfect to five steps (125/125 consume; 82/82 sealed) and **zero for six or more (0/25)** — the exact ceiling V1.7 was built to remove.

5 Phase III (V0.10–V1.2): determinism, integrity, certification

5.1 V0.10 — the deterministic turn (F on its own gate)

A ~0.4 ms deterministic NLP worker replaced the LLM reader on covered domains, and a goal-regression planner replaced depth-limited search (the V0.9 wall autopsied to a `max_depth=5` horizon; backward chaining ran 25 exact-linear expansions at depth 25). Sealed complexity suite: **196/196 across all tiers including forced 20-step chains, at zero LLM calls, identical under llama, haiku, and sonnet managers**; measured energy **0.3 J/correct versus 28,182 for the llama arm (~10^5x)**. A 64-item role-trap benchmark executed zero traps (closing V0.9's 7.95%). The version still reported F: on the combined role+adversarial gate every system exceeded the frozen 2% (best 9.5–10.1%) — "passed every performance bar, failed its own safety bar."

5.2 V0.11 — integrity makes wrongness go to zero (A)

Fact-based semantic state (a six-field identity key: entity/concept/role/temporal/scope/polarity, with an entity graph resolving aliases, ordinals, pronouns, comparatives) plus a **12-check integrity engine** in which only CERTIFIED facts may execute. Sealed results: the unprotected control wrong-executes **44.7%** (semantic) / 40.2% (adversarial); every integrity-bearing system: **0.000, everywhere**. The decomposition is the finding: representation alone adds +43pp accuracy but leaves 6.6% adversarial wrong-fire; checks on a flat state collapse coverage to 3%; only representation *and* integrity give both. The parser gained +39.5pp (44.7→84.2% sealed at zero wrong), a 25,000-query traffic curve came out *exactly flat* (846/1000 at every checkpoint, falsifying two learning hypotheses with mechanism), and measured energy was 0.53 vs 0.75 J/query with an **84.1% workload-level saving** — the escalated 15.8% of queries carried 99.7% of residual energy. Coverage became, explicitly, the energy frontier.

5.3 V1.0 — the integrated engine and the acquisition channel (C)

A fresh repository (`reflex-engine`) integrated the arc: a gateway returning CERTIFIED_REFLEX / LLM_ESCALATED / ABSTAINED with machine-readable certificates; a physics pack of 77 kinds and 58 capabilities loaded by injection (13/13 byte-identical); full parity with the frozen V0.11 engine (five suites, zero item-level diffs, three times; 196/196 at zero LLM calls; 0.403 J/query). The acquisition experiment put breeder and model-miners behind one admission gauntlet on a 5,000-item pure-residual stream: the local breeder was **inert** (0 candidates), llama-8B mined 132 active reflexes with **zero lift**, while **haiku added +5.8pp (p=3.7e-9) and sonnet +7.2pp (p≈0) of certified sealed-residual coverage at wrong-fire 0.000 in every arm, probe, and final** — coverage equalled accuracy. Ablations completed the picture: integrity off → 6.6% adversarial wrong-fire; pure-LLM arms 24.6–36.9% wrong (and the measured 1,910 J/query llama baseline); gauntlet off → still 0 wrong-fire but polluted coverage — defense in depth, each layer catching a different failure. Sonnet's mining cost ~$3.5 for the +7.2pp.

5.4 V1.1–V1.2 — scale-up, then formal certification (A, then B)

V1.1 closed its generator universe (parser + out-of-family 100.0%; a 180,000-item sweep at coverage 1.0000, zero wrong, zero residual), introduced typed reflexes with per-type gates, and ran a 100,000-query commercial simulation: **95.5% certified, zero wrong, 20 reflexes minted live**, with the mint ranking opus 12 > haiku 8 > sonnet 6. One latent defect (an ordinal-reference gap) wrong-fired 1/76 on a fresh seed and was fixed and superseded — "seed luck is not coverage" entered the arc's vocabulary. V1.2 then certified the machine formally: a **seed sweep of 10,160 adversarial traps across 65 seeds at 0.000 wrong-fire**, checked by an independent `certify.py` and issued as a one-page certificate; the energy figure was corrected by batch metering to **21 mJ per certified serve** (the earlier 1.93 J proved contamination; 0.85 ms/query compute). V1.2 also made first contact with real language — and the certified engine abstained on **519 of 520** story problems, with sealed acquisition lift *zero* in both arms. The gap was named precisely: not vocabulary but *story composition*; and a first scale inversion appeared (opus capability-mining produced 17 wrong candidates against sonnet's 7) whose diagnosis — "dimensional gates certify physics, not story semantics" — became V1.3's problem statement.

6 Phase IV (V1.3–V1.5): the real-language assault, and why it failed

Three experiments attacked open story problems from three directions. Their combined verdict caused the pivot; each also left a permanent positive asset.

6.1 V1.3 — the model reads, the engine computes (D)

The parse/verify split: the model emits a typed plan IR; the engine verifies (seven checks) and computes; no LLM number enters certified state. On a sealed 300-item GSM8K/ASDiv sample: sonnet **209 certified / 3 wrong (1.44%) at 69.7% coverage** — real language finally read, against V1.2's 519/520 abstentions — while haiku's 1/38 = 2.63% breached the frozen 2% gate, forcing outcome D over aggregate readings of A. Permanent assets: the first certified **non-arithmetic** track (provenance-certified retrieval: 289 served, 0 wrong, two corpora; two disagreeing live sources *abstain*, never vote); and the R1/R2/R3 comparison showing that injecting certified values into a model's context is **worse than plain RAG (57.3% vs 36.8% wrong)** — the guarantee belongs to *whoever computes*, not to the context window. Permanent warnings: stored plan reflexes replayed positionally served **7/7 wrong** ("checks verify a plan matches the TEXT, never that it answers the QUESTION"); and five wrong answers passed every check *with cross-model corroboration* — two model families produced the identical misreading of "three times more", because interpretation priors are shared. Resampling does not recover semantics.

6.2 V1.4 — the representation was never the hard part (D)

A relational state substrate with 15 integrity classes took deterministic semantic-error detection from 10.0% to **100% at 0% false positives** on 600 sealed corrupt/correct pairs, made the canonical error 90–(49–19) *unrepresentable*, and refused V1.3's plan-reuse failure 6/6. Then the same machinery met 240 sealed real-language items and certified **zero** in all four arms — the producers could not *make* the representation: two samples of one model produced structurally different states ~3/4 of the time (agreement blocked 68.3% of items). The lesson, verbatim from the report: *the bottleneck moved from checking the reading to producing it; V1.4 is safe and inert, V1.3 useful and imperfect, and nothing yet both.* The version also measured the general-purpose configuration's economics end-to-end — an LLM-in-the-loop state producer answering 0/60 while spending 19.9x the GPU energy of the direct LLM — removing any economic reason to persist with that configuration and adding a second driver to the pivot.

6.3 V1.5 — composition, named and measured (E)

A 6 ms deterministic reader extracted quantities with zero unverifiable spans and named its missing field 95%+ of the time — and the full pipeline certified **5 of 800 sealed items (0.63% coverage), 2 of them wrong (40% wrong-fire)**. Four composition guards took dev wrong-fire 43.75%→0%; sealed stayed 40% — *guards encode which compositions are wrong, never which is right*. A naming fallback assigned the kind `payback_months` 131 times to snow depths and schoolchildren — worse than UNKNOWN, because a fabricated kind carries a dimension and *passes* the dimensional gate. The report's closing recommendation, quoted: three versions from three directions now say the deterministic path cannot answer open stories safely at useful coverage; the asset that survived every version is the **abstention**.

6.4 The pivot decision

The evidence, in one place: V1.3 read language at 69.7% coverage but 1.44% wrong (and its best-behaved track was the *computable* one); V1.4 built a perfect checker that certified 0/240 because reading cannot be reliably produced; V1.5 added the cheapest possible reader and served 5/800 at 40% wrong. Meanwhile every *scientific-register* result in the same period — V1.1's 95.5% at zero wrong over 100k, V1.2's 0.000 over 10,160 traps, V1.0's zero-wrong-fire acquisition — kept holding. The programme therefore scoped, deliberately and on the record, to the register where answers are *computable* and witnesses are *independent*: the scientific/arithmetic/logical reflex.

7 The pivot lands: V1.6, the scientific reflex engine (B)

V1.6 rebuilt the question: not “can the engine read stories?” but “between V1.1's 100% on its own grammar and V1.2's 0% on stories, where does *foreign-phrased calculator register* sit — and can a model-grown library move it?” Four frontier models (haiku, sonnet, opus, and fable) minted parse procedures against a 47-leaf topic taxonomy; every candidate faced an mpmath-only independent witness (deliberately *not* sympy, breaking shared-substrate circularity) and probation with one-strike admission.

- **Sealed, twelve runs, 118k+ serves: wrong-fire 0.0 everywhere.** Mixed sealed traffic: baseline 45.7% → 70.35% with the union library (+24.6pp against a 5pp bar). Foreign-phrased sealed: **35.4%** → **55.5%** with 271 admitted reflexes (bar 40%). Dev→sealed held (52.8→55.5) — the V1.5 guard collapse did not recur, because reflexes are *parse procedures behind an independent witness*, not composition rules.
- **Scale inversion falsified for good.** Probation wrong-kills fell monotonically with model scale — 16 (haiku) / 3 (sonnet) / 1 (opus) / 1 (fable) — and sealed foreign lift rose (8.9/11.2/16.6/13.7pp). V1.1's inversion was architectural, not a law of models: when producers propose *procedures* and every output passes an independent witness, fluency becomes coverage. Opus independently declined to mint a family that “would certify a different question” and flagged its own g=9.81 assumption — UNKNOWN>WRONG, arrived at by a miner.
- **Depth was already non-monotone** — d1 87.5 / d2 50.0 / d3 99.7 / d4 0-by-guard, zero wrong at every rung: *shape*, not depth, ends coverage. This observation seeded V1.7.
- **A verifier honestly failed its bar**: impossibility-style checks recalled only 3.0–6.7% of escalated-share errors (bar 10%) at zero false fires — computational slips are invisible to impossibility checks; catching them requires recomputation, which is what the engine already does on everything it serves. Measured LLM wrong-rates on the escalated share — haiku 33%, sonnet/opus/fable 11–12% — are the arc's best argument for certified serving: what the engine escalates is what LLMs get wrong most.

8 Composition at million scale: V1.7 in brief (E)

V1.7 (reported in full in the companion paper *Certified Multi-Reflex Chains at Million Scale*) asked whether the pivot's skills cohere. On a hash-sealed corpus of 1,000,000 generated chain questions (depths 2–6; 60.6% non-monotonous; 71.7% cross-domain; 3,624 never-answer distractors):

- **Deterministic cohesion.** A role-aware composer + gate (schema/DAG, type-and-unit flow, ask-target, stated-number honesty) + dual sympy/mpmath execution served **112,170 sealed chain answers with zero errors** — 30.4% of the 3+-step stratum, 17.4% even at six chained steps. The V0.9 depth wall is gone.
- **Model planning.** Structuring 1,200 sealed escalations each: opus 1,150 / fable 1,138 / sonnet 1,118 certified with **zero wrong**; haiku 1,077 with 13 wrong (all one dual-role mechanism). Structure-finding took 4.60–6.45 s median and was **near-flat in depth** (+10–19%, k=3→k=6); planning lifts the 3+-step stratum to an estimated 95.2–97.1%.
- **The negative result.** Reusing certified plans as positional regex templates produced **8,842 wrong answers across 4,000,000 evaluations** (sealed wrong-fire 0.245–0.800% per arm, monotone improving with scale), while the role-aware composer inside identical runs made zero errors. Every wrong traced to role-blind number binding; a literal-presence fix closed only the donor-constant subset (0.73→0.46% on a dev probe). Role-aware reuse is V1.8's pre-registered problem statement.

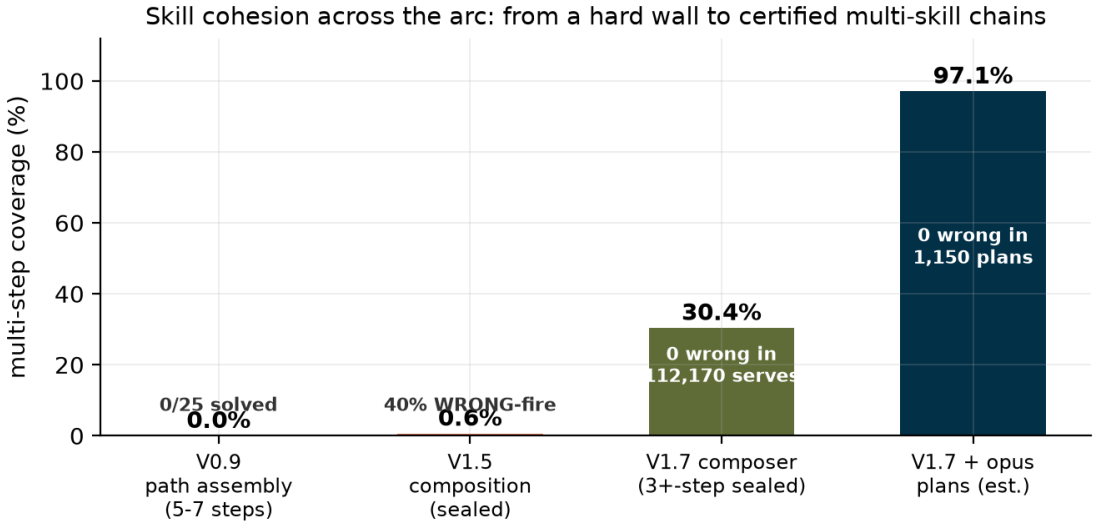


Figure 2: The composition thread across the arc. V0.9: 0/25 beyond five steps. V1.5: 0.63% coverage at 40% wrong-fire. V1.7: 30.4% deterministic at zero wrong over 112,170 sealed serves; ~97% with certified opus planning at zero wrong (estimate from a 1,200-item uniform sample of the escalation pool).

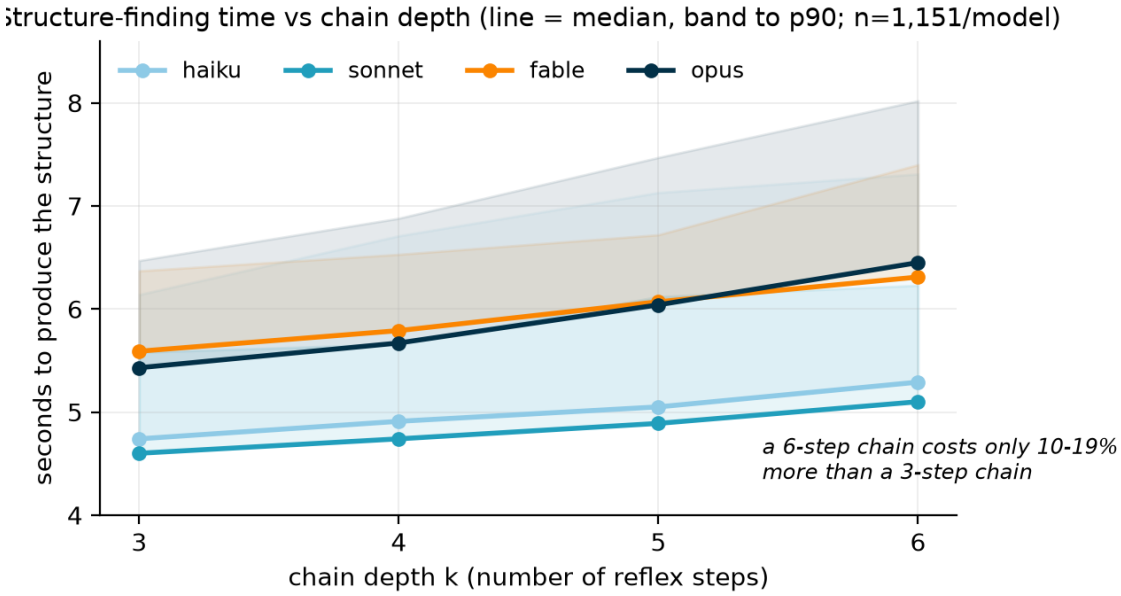


Figure 3: V1.7 structure-finding time versus chain depth (median; band to p90; n=1,151 per model, harness-observed). Depth is nearly free; the fastest planners include the only unsafe one.

9 What nineteen experiments agree on

9.1 The independent-witness principle

Every zero-wrong record in the arc sits behind an independent recomputation: V0.3's typed preconditions, V0.8's typed-expression contract, V0.11's integrity engine, V1.6's mpmath witness, V1.7's dual execution. Conversely, every mechanism that certified on *agreement or plausibility* eventually served nonsense: cross-model corroboration reproduced the identical misreading (V1.3), impossibility checks missed computational slips (V1.6), and a fabricated kind with a dimension sailed through a dimensional gate (V1.5). Six “green checks that could not go red” later, the rule is absolute: **certify by recomputation, and give every checker a positive control.**

9.2 Models propose; engines compute

The division of labour discovered in V0.4 held under every subsequent test. Where models supply *values*: 66.1% (V0.4), 36–39% with 104–117 wrong numerics (V0.8), 24.6–36.9% wrong (V1.0 pure-LLM arms). Where models supply *structure* the engine then computes and witnesses: 84.3% (V0.4), 94.9–95.4% (V0.8), +5.8/+7.2pp at 0.000 wrong-fire (V1.0), 271 admitted reflexes at zero sealed wrong-fire (V1.6), 3,406 certified sealed plans at zero wrong (V1.7, three larger models). The one place structure-supply failed — positional plan replay (V1.3's 7/7; V1.7's 8,842) — is precisely where a *stored* structure was rebound to a new question without re-deriving roles.

9.3 Scale is monotone once the architecture is right

Under value-handoff and free-form architectures, bigger models were not reliably safer (V1.1: opus minted most; V1.2: opus minted *most wrong*). Under procedure-plus-witness architectures every scale gradient in the arc is monotone: V1.6 probation wrong-kills 16/3/1/1 and foreign lift 8.9/11.2/16.6/13.7pp; V1.7 sealed coverage 39.2/44.6/47.2/48.8%, wrong-fire 0.800/0.631/0.384/0.245%, template error 18.7/4.0/1.9/1.06%, sealed plan wrongs 13/0/0/0 (haiku→opus ordering; fable between sonnet and opus throughout). **Architecture decides whether scale buys safety.**

9.4 The semantics ladder — what checks cannot see

One lesson recurred at every level of abstraction, each time one layer up: dimensional gates certify physics, not story semantics (V1.2); parse checks verify text-consistency, not question-answering (V1.3); resampling varies the plan, never the reading (V1.3); guards encode wrong compositions, not right ones (V1.5); magnitude-level honesty is not role-level honesty (V1.7). The programme's answer is not ever-stronger filters — it is moving the semantic burden into *typed, executable structure* whose meaning the engine can recompute (the composer), and treating everything else as UNKNOWN.

9.5 Energy: the scientific-solver position

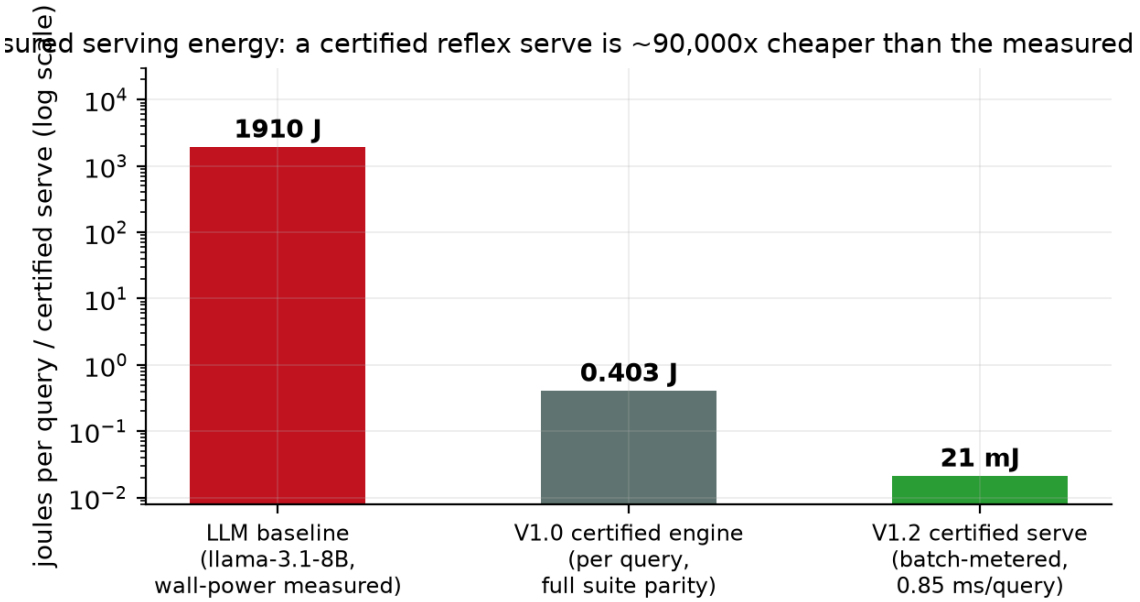


Figure 4: Measured serving-side energy of the certified engine (log scale; wall-power, same workstation). The 21 mJ figure is batch-metered (V1.2) after instrument audit B24; the LLM baseline is the measured llama-3.1-8B arm (V1.0). Configurations that place a model inside the serving loop are out of scope since the pivot and are not charted.

Component-level, the position is measured: **21 mJ and 0.85 ms per certified serve against 1,910 J for the measured LLM baseline (~9x10^4)**, with workload-level savings of 84.1% (V0.11) and 95.4% (V1.1's simulated commercial mix) when the engine fronts traffic, and residual energy concentrating in the escalated share (99.7% in 15.8% of queries — coverage *is* the energy frontier). Since the pivot, no model sits in the serving loop at all: V1.7's 4,000,000 evaluations completed with zero model calls, models acting only offline as amortized proposers (48–75 serves per admitted entry on replay traffic). A fleet-level energy measurement of the V1.7 configuration remains registered future work; the arc's instrumentation history (B7, B24) is the reason such a number is withheld until measured.

10 Standing and outlook

Against the programme goal — **AI scientific solvers that consume orders of magnitude less serving energy, with learned skills that cohere** — nineteen experiments have produced: a certified deterministic core with no wrong serve on sealed data since the pivot (V1.6's 118k+ and V1.7's 112k+ serves, plus ~550k across integrity-era suites); a skill-acquisition channel proven across four frontier models with quality monotone in scale; skills that compose into six-step certified chains; a quantified planning interface (~5 s per novel structure, near-flat in depth); and serving economics measured at the millijoule level. The open items are equally explicit: role-aware reuse (V1.8's pre-registered bet: the 8,842-wrong class vanishes at unchanged coverage when templates bind by role), a fleet-level energy measurement of the lean configuration, coverage growth at depth (17.4% at k=6), and — deliberately deferred, not abandoned — the open-language frontier, to be re-entered only with the assets this arc proved: independent witnesses, typed structure, and an engine that knows what it does not know.

The journey in one sentence: the programme set out to make answers cheap, learned through three expensive failures that the only answers worth serving are the ones a machine can recompute, and found — on the scientific register that admits recomputation — that certified skills can be learned from frontier models, composed into chains, and served by the hundred thousand without a single wrong answer.

Appendix A The arc, version by version

ver	outcome	headline measured result	lesson carried forward
V0.1	PoC	hybrid 85.5% vs LLM 77.0% @ 0.19x FLOPs; reflex ~2,950x cheaper	recognition is the floor cost; neural reflexes hallucinate
V0.2	B	graph 87.5% @39% LLM vs 75.7% direct; hashed router 200,000x cheaper at 1e6	graph capability; embedder removable
V0.3	A	100% on 62 unseen typed compositions vs 38.7% LLM; 0 incorrect reflex	typed preconditions turn errors into abstentions
V0.4	B	value handoff 66.1% (harmful); expression handoff 84.3% (+14.9pp p=0.008)	models propose structure; engines compute values
V0.5	A	77.7% @ 574 J/correct vs 68.6% @ 1,687; breeder 86.0% @ 321	external OOV 8%: parser-bound
V0.6	B	coverage 0->57% live; 0 wrong numerics through 50% pollution; hybrid 81.8% vs 69.4%	learning: pollution hits recall, not correctness
V0.7	E	substrate transfer 0->55% held-out (0 wrong); OOV <45% at every model scale	recognition wall is scale-independent
V0.8	C	recognition 83.1% @ 0.29% wf; PM 95.4% vs 5.6%; raw-value 36-39% vs typed 94.9%	types decide safety; the index only organizes
V0.9	F	SQR external 94% vs 42%; depth wall: 125/125 at <=5 steps, 0/25 at 6+	the wall breaks by representation; depth by search horizon
V0.10	F	196/196 incl. 20-step @ 0 LLM calls; 0.3 vs 28,182 J/correct (~1e5x); gate failed	graded regression removes the depth wall
V0.11	A	wrong-exec 44.7% -> 0.000 everywhere; parser +39.5pp @ 0 wrong; 84.1% vs 44.6%	verification buys accuracy; integrity removes wrongness
V0.12	C	haiku +5.8pp / sonnet +7.2pp certified @ 0.000 wf; llama miner 0 lift; 1,910 traps	acquisition measured, proven; defense in depth
V1.1	A*	100k sim: 95.5% certified, 0 wrong, 20 minted live; 180k sweep 0 residual	seed luck is not coverage (B2: 1/76 on fresh seed)
V1.2	B	10,160 traps x 65 seeds @ 0.000 wf, formal certificate; 21 mJ/serve; stories 519/520	claims certify physics, not story semantics
V1.3	D	sonnet 209 cert / 3 wrong (1.44%) @ 69.7% cov; retrieval 289 @ 0 wrong	injection 7/7 wrong; CAG corroboration shares reading priors
V1.4	D	detection 10% -> 100% @ 0% FP (600 sealed); real language: 0/240 certified	bottleneck moved from checking to producing the reading
V1.5	E	5/800 certified, 2 wrong (40% wf); guards dev 43.75->0%, sealed 40%	composition is the bottleneck; the abstention is the asset
V1.6	B	PIVOT: foreign 35.4 -> 55.5% @ 0.0 wf; 271 reflexes; 118k+ sealed serves, 12 dumps	procedures + independent witness make scale monotone
V1.7	E	112,170 sealed chain serves @ 0 wrong; 3,406 sealed plans @ 0 wrong (3 breaks); composed 8,848 wrong in 40;	role-blind ones always did

Table A1: All nineteen experiments. A* = outcome A with one reported protocol violation. Outcomes are the pre-registered ladder letters; several “E/F” versions contain the arc's most load-bearing positive results, and several “A/B” versions its sharpest warnings — the letter grades safety compliance, not usefulness.

Reproducibility. Every version is git-tagged with a read-only frozen snapshot (`reflex-poc-v0.x-frozen`, `reflex-engine-v1.x-frozen`), a per-version report (MD, and PDF from V0.8), a numbered bug audit, and a single-command reproduction (`run_v0_x_experiment.py` / `v1x_run.py`). Sealed corpora are committed with SHA-256 raw-byte hashes. Local model: llama-3.1-8B (ollama, q4_K_M) throughout V0.5–V1.0 energy work; frontier models accessed as isolated bridge agents. The V1.7 companion paper (*Certified Multi-Reflex Chains at Million Scale*, same repository) carries the full V1.7 tables, figures, and failure anatomy.